

Visualization - Practical Work

Data Visualization

Statement

- Available in Moodle
- Read it carefully!



DV 2026/2027

Data Visualization: Visualization Project

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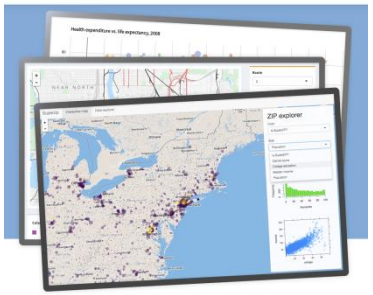
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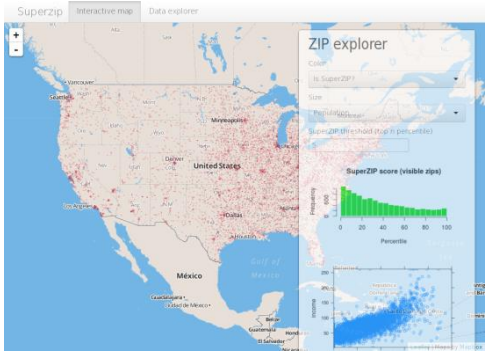
Goal

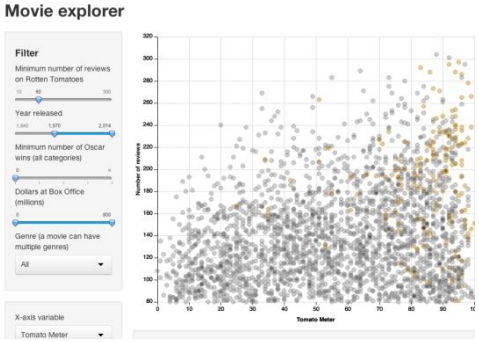
- Put into practice the concepts learnt during the theory classes:
  - Visualization project methodology
    - Task abstraction
    - Data abstraction
    - Design: encoding theory
  - Practical implementation
    - R + Shiny
    - Python + Shiny
    - Js + D3 + Observable

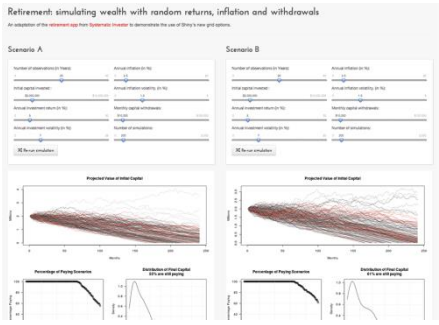
Shiny



- Shiny is an R package
- Also available for Python
- Great for prototyping! (and dashboards)
- Visualization + R Data Analysis → Visual Analytics
- Reactive programming
- Examples of what can be done: https://shiny.posit.co/r/gallery/









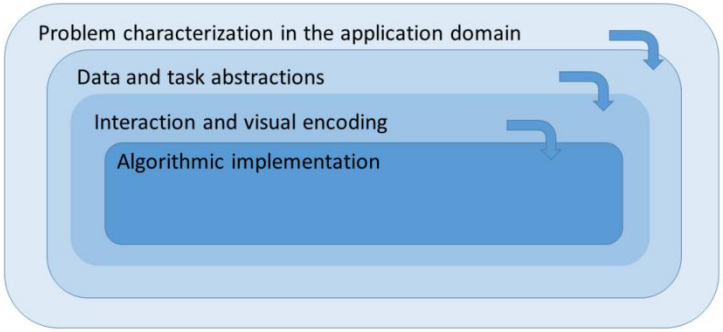
Visualization Project

- **Design and implement your own interactive visualization analysis tool**

- Key aspects:
  - Analysis tasks (choose a dataset)
  - Methodology:
    - Characterize in the problem domain
    - Data and task abstraction
    - Visualization design
    - Implementation

Vis. Project: methodology

- Imagine you are working in a real-life environment:
  - Users/analysis: domain experts
  - Vis designers
  - Vis developers

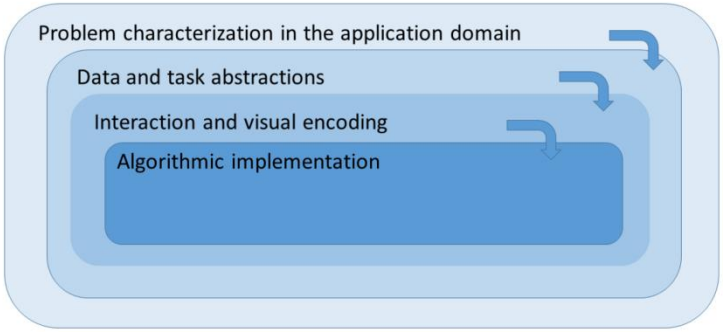


- Select a domain:
  - Choose a dataset (I encourage you to look for something that is of your interest!)
  - Think about the questions an analyst would like to answer with the data

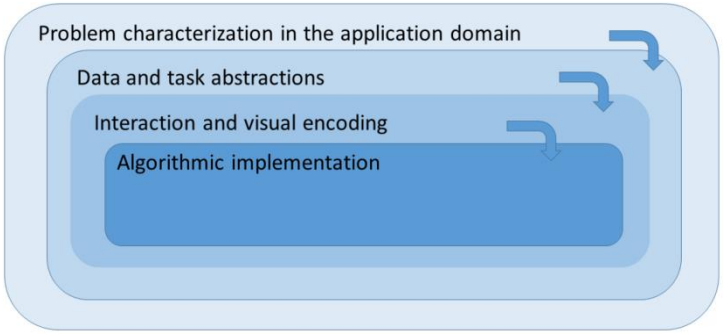
Vis. Project: methodology

- Data and task abstraction → domain-independence
  - Describe the data-set and its attributes in an abstract way
  - Describe the analysis questions with abstract tasks

- Decide which idioms would help answer those questions effectively:
  - Visual encoding
  - Interaction



Vis. Project: evaluation



- Mandatory:
  - Follow the methodology shown in class
    - Choose a data-set
    - Pose analysis questions
    - Data and task abstractions
  - 3 idioms (4 for groups of 5)
  - Meaningful interaction in each idiom (filtering, aggregation, selection, …)
  - Publish it online and provide a link in the report
    - Shinyapps.io, Posit Connect Cloud, Observable
  - Report: brief and concise but comprehensive
    - Structure it following the design abstraction levels
    - Screenshots that show how the tool helps answering the analysts' questions
- Extra:
  - Data analysis algorithms integration
  - Multiple coordinated views
  - Idioms with specific arrangements (if justified!)

Common mistakes to avoid

- Begin the project programming 0:)
- Choosing a data-set too complex or too simple
- Questions not well suited for visualization
- Using non-justified arrangements
  - 3D
  - Maps

Teacher's role

- I will review with you at least the following steps:
  - Discuss the chosen data-set (mandatory!)
  - Discuss the questions posed (very recommended)

- This can be done using:
  - The dedicated time in class
  - MS Teams
  - Email

Deadlines

- Approval of the chosen data-set:
  - Oct 5th, 2026
  - During time class (preferred) / Teams / Email
- Project delivery deadline:
  - Jan 11th, 2027 (Exam date)
  - Delivery through Moodle

Group work

- Done in groups of 4 or 5 people
- Create your group in Moodle
- If you can not find partners:
  - Please let me know
  - Forum in Moodle for contacting other students
  - Join a group that is not complete